

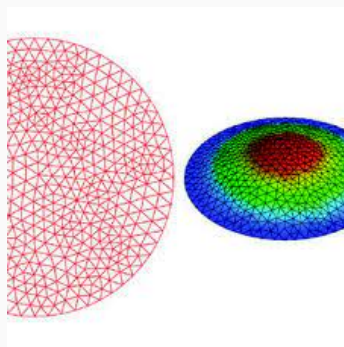
Dynamic analysis of Kirchhoff plates

Special structural elements

ME473 Dynamic finite element analysis of structures

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2025



Where do we stand?

Week	Module	Lecture topic	Mini-projects
1	Linear elastodynamics	Strong and weak forms	
2		Galerkin method	
3		Finite element method	Groups formation
4		Systematization of the procedure	Project 1 statement
5		3d elements, numerical integration	
6	Special structural elements	Bars and trusses	
7		Planar beams	Project 1 submission
8		Frames and grids	Project 2 statement
9		Kirchhoff-Love plates	

Summary

- Recap weeks 6 to 8
- Kirchhoff-Love plate theory
- AMC thin plate bending elements
- CR thin plate bending elements

Recommended readings

- (P) Petyt, Introduction to finite element vibration analysis (chap. 6)
- (N) Neto et al., Engineering Computation of Structures (chap. 6.1)
- (O) Ochsner, PDE for classical structural members (chap. 6)
- (G) Gmür, Méthode des éléments finis (chap. 3)

Recap weeks 6 to 8

Vibrations of unidimensional structures

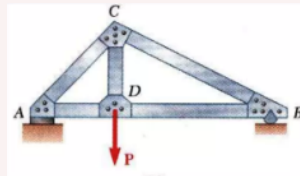
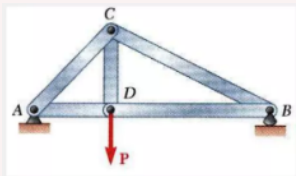
Trusses vs frames

Truss

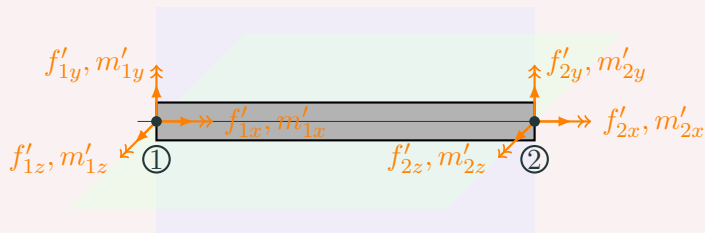
- Structure composed of oriented bar (rod) elements, connected by frictionless pins, carrying **axial forces** only.

Frame

- Structure composed of oriented beam elements, connected by welding, carrying **transversal, axial forces and torsion**.



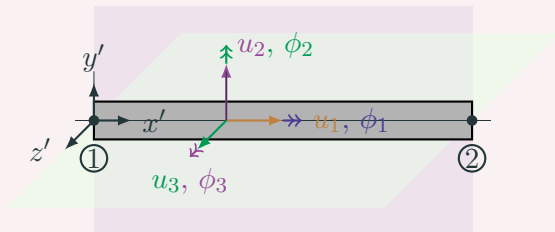
Three-dimensional thin beam structure



Three-dimensional thin beams are uniaxial (slender) element that can support:

- axial loads f'_{ix} ,
- torsional loads m'_{ix} ,
- bending in the $x' - y'$ plane: f'_{iy} and m'_{iz} ,
- bending in the $x' - z'$ plane: f'_{iz} and m'_{iy} .

Differential equations governing the dynamics of thin beams



Bars

$$EA \partial_{x'x'}^2 u_1(x', t) = \rho A \ddot{u}_1(x', t)$$

Shafts

$$GJ \partial_{x'x'}^2 \phi_1(x', t) = \rho J \ddot{\phi}_1(x', t)$$

Planar ($Ox'y'$) beams

$$\partial_{x'x'}^2 (EI_{z'} \partial_{x'x'}^2 u_2(x', t)) + \rho A \ddot{u}_2(x', t) = 0$$

Planar ($Ox'z'$) beams

$$\partial_{x'x'}^2 (EI_{y'} \partial_{x'x'}^2 u_3(x', t)) + \rho A \ddot{u}_3(x', t) = 0$$

$I_{y'}$ and $I_{z'}$ are the cross-sectional moments of inertia with respect to the axes y and z .

Displacements discretization

Total of six nodal displacements at each unconstrained node:

- three translation components q'_{ix} , q'_{iy} and q'_{iz} along the x , y , z axes, and
- three rotational components about these axes ϕ'_{ix} , ϕ'_{iy} and ϕ'_{iz} .

$$\mathbf{q}_{loc}(t) = \begin{bmatrix} q'_{1x}(t) \\ q'_{1y}(t) \\ q'_{1z}(t) \\ \phi'_{1x}(t) \\ \phi'_{1y}(t) \\ \phi'_{1z}(t) \\ q'_{2x}(t) \\ q'_{2y}(t) \\ q'_{2z}(t) \\ \phi'_{2x}(t) \\ \phi'_{2y}(t) \\ \phi'_{2z}(t) \end{bmatrix}$$

$$u^h(x', t) = \mathbf{H}(x') \mathbf{q}_{loc}(t)$$

$$h_1(x') = 1 - x'/\ell$$

$$h_2(x') = 2(x'/\ell)^3 - 3(x'/\ell)^2 + 1$$

$$h_3(x') = 2(x'/\ell)^3 - 3(x'/\ell)^2 + 1$$

$$h_4(x') = 1 - x'/\ell$$

$$h_5(x') = x'(1 - x'/\ell)^2$$

$$h_6(x') = x'(1 - x'/\ell)^2$$

$$h_7(x') = x'/\ell$$

$$h_8(x') = 3(x'/\ell)^2 - 2(x'/\ell)^3$$

$$h_9(x') = 3(x'/\ell)^2 - 2(x'/\ell)^3$$

$$h_{10}(x') = x'/\ell$$

$$h_{11}(x') = x'(x'/\ell)(x'/\ell - 1)$$

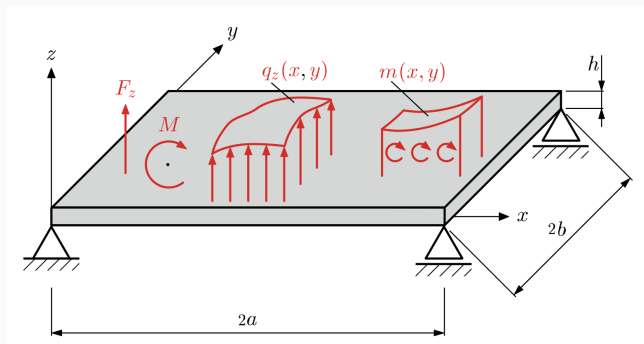
$$h_{12}(x') = x'(x'/\ell)(x'/\ell - 1)$$

- The stiffness matrix $\mathbf{K}_{loc}$ and mass matrices $\mathbf{M}_{loc}$ for a three-dimensional beam element are written as the superposition of the **axial** matrix, the **torsional** matrix and the **flexural** matrices in the two bending planes.

Classic plate theory

Plate structure

- Plate structures are geometrically similar to structures of the 2D plane stress problem, but it usually carries only transversal loads that lead to **bending** deformation of the plate.
- For example: floors of a building, aerospace and ships structures, etc...



(Credit: (O))

Plate models

Kirchhoff (1888) and Love (1945)

- *Shear free plates*: thin plates where the contribution of shear force on the deformations is neglected.
- Two-dimensional extension of the Bernoulli-Euler beam theory.

Mindlin (1951) and Reissner (1945)

- *Shear deformable plates*: thick plates where the contribution of shear force on deformations is considered.
- Two-dimensional extension of the Timoshenko beam theory.

Gustav Kirchhoff



1888

Augustus Love
Eric Reissner



1945

Raymond Mindlin



1951

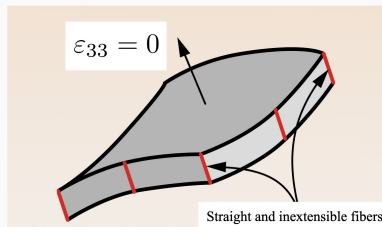
Geometry assumptions

- The thickness of the plate h is constant and much smaller than the planar dimensions a and b :

$$h/a < 0.1 \quad \text{and} \quad h/b < 0.1.$$

- **Inextensibility of transverse fibers:** transverse normal strain is negligible.

$$\varepsilon_{33} = 0.$$



(Credit: (G))

Material and loads assumptions

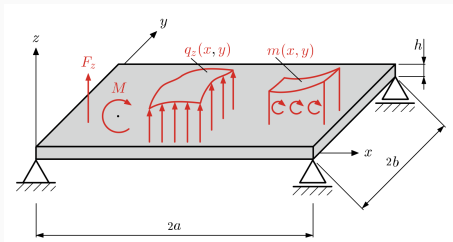
Material

- The material is homogenous and linear-elastic according to Hooke's law for a plane stress state

$$\sigma_{33} = \sigma_{13} = \sigma_{23} = 0.$$

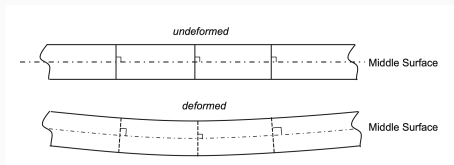
Loads

- External forces act only perpendicular to the $x - y$ plane, the vector of external moments lies within the $x - y$ plane.
- Displacement $u_3(x, y, t)$ is small compared to h : $u_3 \lesssim 0.2h$.



Kirchhoff assumption

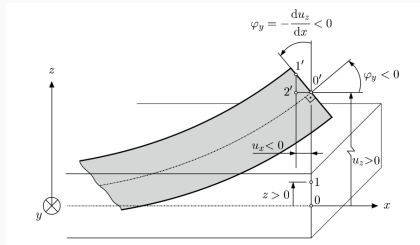
Rectilinearity of the normals: Bernoulli's hypothesis is valid, i.e. a cross-sectional plane stays plane and unwrapped in the deformed state. This means that the shear strains ε_{13} and ε_{23} due to the distributed shear forces q_x and q_y are neglected.



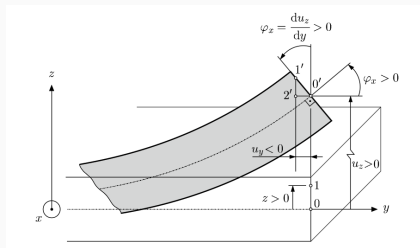
A straight fiber that is perpendicular to the middle plane of the plate before deformation remain straight and normal to it after deformation.

(Credit: (N))

Kinematics assumptions



$$\begin{aligned} -\varphi_2 &\approx \sin(-\varphi_2) = -\frac{u_1}{z} \\ -\varphi_2 &\approx \tan(-\varphi_2) = \frac{du_3}{dx} \\ \Rightarrow u_1 &= -z \frac{du_3}{dx} \end{aligned}$$



$$\begin{aligned} \varphi_1 &\approx \sin(\varphi_1) = -\frac{u_2}{z} \\ \varphi_1 &\approx \tan(\varphi_1) = \frac{du_3}{dy} \\ \Rightarrow u_2 &= -z \frac{du_3}{dy} \end{aligned}$$

Transverse displacement u_3 is the *only* independent variable:

$$\mathbf{u} = \begin{bmatrix} -z \frac{\partial u_3}{\partial x} \\ -z \frac{\partial u_3}{\partial y} \\ u_3 \end{bmatrix}$$

Deformation is exaggerated in the figures for better illustration.

Strain-displacement relation

- Using classical engineering definitions of strain:

$$\varepsilon_{ii} = \partial_i u_i \quad \text{and} \quad \gamma_{ij} = \partial_i u_j + \partial_j u_i$$

we obtain

$$\underbrace{\begin{bmatrix} \varepsilon_{11} \\ \varepsilon_{22} \\ \gamma_{12} \end{bmatrix}}_{\boldsymbol{\varepsilon}} = -z \underbrace{\begin{bmatrix} \partial_{xx}^2 \\ \partial_{yy}^2 \\ 2\partial_{xy}^2 \end{bmatrix}}_{\nabla_k} u_3 = z \underbrace{\begin{bmatrix} \kappa_x \\ \kappa_y \\ \kappa_{xy} \end{bmatrix}}_{\boldsymbol{\kappa}}$$

- $\boldsymbol{\kappa}$ is the matrix that contains the changes in the **curvature of the plate**, given as $\boldsymbol{\kappa} = -\nabla_k u_3$.
- Note that $\varepsilon_{12} = \varepsilon_{23} = 0$ due to Kirchhoff assumptions and $\varepsilon_{33} = 0$ due to the inextensibility of transverse fibers assumption.

Constitutive equation for isotropic material

- Classical plate theory assumes a plane stress state:

$$\sigma_{33} = \sigma_{13} = \sigma_{23} = 0.$$

- Constitutive equation for *isotropic* material is $\boldsymbol{\sigma} = \mathbf{C}\boldsymbol{\varepsilon}$ or $\boldsymbol{\varepsilon} = \mathbf{D}\boldsymbol{\sigma}$ where

$$\begin{bmatrix} \sigma_{11} \\ \sigma_{22} \\ \sigma_{12} \end{bmatrix} = \frac{E}{1 - \nu^2} \begin{bmatrix} 1 & \nu & 0 \\ \nu & 1 & 0 \\ 0 & 0 & \frac{1-\nu}{2} \end{bmatrix} \begin{bmatrix} \varepsilon_{11} \\ \varepsilon_{22} \\ \gamma_{12} \end{bmatrix},$$

or

$$\begin{bmatrix} \varepsilon_{11} \\ \varepsilon_{22} \\ \gamma_{12} \end{bmatrix} = \frac{1}{E} \begin{bmatrix} 1 & -\nu & 0 \\ -\nu & 1 & 0 \\ 0 & 0 & 2(\nu + 1) \end{bmatrix} \begin{bmatrix} \sigma_{11} \\ \sigma_{22} \\ \sigma_{12} \end{bmatrix}.$$

- $\mathbf{C}$ is the *elasticity matrix* and $\mathbf{D} = \mathbf{C}^{-1}$ is the *elastic compliance matrix*.

Constitutive equation for orthotropic material

- The constitutive equation for *orthotropic* material is $\boldsymbol{\sigma} = \mathbf{C}\boldsymbol{\varepsilon}$ or $\boldsymbol{\varepsilon} = \mathbf{D}\boldsymbol{\sigma}$ where

$$\begin{bmatrix} \sigma_{11} \\ \sigma_{22} \\ \sigma_{12} \end{bmatrix} = \begin{bmatrix} Q_{11} & Q_{12} & 0 \\ Q_{12} & Q_{22} & 0 \\ 0 & 0 & Q_{33} \end{bmatrix} \begin{bmatrix} \varepsilon_{11} \\ \varepsilon_{22} \\ \gamma_{12} \end{bmatrix}$$

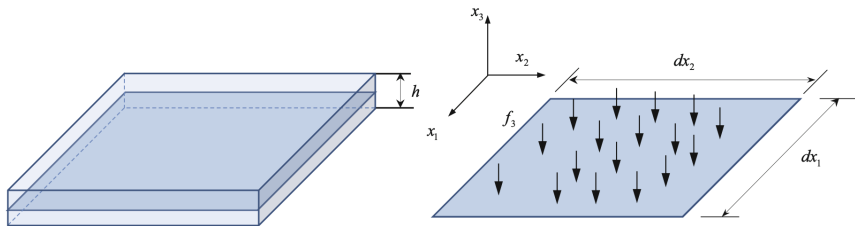
where

$$Q_{11} = \frac{E_1}{1 - \nu_{12}\nu_{21}}, \quad Q_{12} = \frac{\nu_{12}E_2}{1 - \nu_{12}\nu_{21}}, \quad Q_{22} = \frac{E_2}{1 - \nu_{12}\nu_{21}}, \quad Q_{33} = G_{12}$$

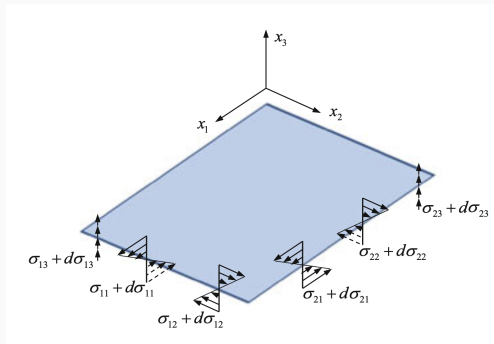
- The orthotropic properties of the lamina are given: E_1 , E_2 , ν_{12} , G_{12} and $\nu_{21} = \nu_{12}E_2/E_1$ applies.

External forces

Consider a plate cell of dimensions $dx_1 \times dx_2 \times h$ that is submitted to external forces, here denoted by f_3 , and inertial force proportional to the material density.



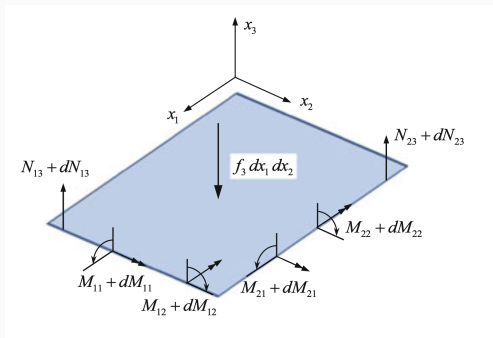
Distributed normal and shear stresses



Normal and shear stresses distributions through the thickness of the plate element:

- linear distributed normal stresses σ_{11} and σ_{22} ,
- linear distributed shear stresses σ_{12} and σ_{21} ,
- parabolic distributed shear stresses σ_{23} and σ_{13} .

Moments and shear forces



Moments and shear forces acting along the edge of the plate:

- bending moments M_{11} and M_{22} ,
- twisting moment M_{12} ,
- shear forces N_{13} and N_{23} .

$$\mathbf{M} = \begin{bmatrix} M_{11} \\ M_{22} \\ M_{12} \end{bmatrix} = \int_{-\frac{h}{2}}^{\frac{h}{2}} x_3 \begin{bmatrix} \sigma_{11} \\ \sigma_{22} \\ \sigma_{12} \end{bmatrix} dx_3 = \int_{-\frac{h}{2}}^{\frac{h}{2}} x_3 \boldsymbol{\sigma} dx_3 = -\mathbf{C} \nabla_k u_3 \int_{-\frac{h}{2}}^{\frac{h}{2}} x_3^2 dx_3 = -\frac{h^3}{12} \mathbf{C} \nabla_k u_3$$

Dynamic equilibrium equation

- Equilibrium condition for the vertical forces:

$$\frac{\partial N_{13}}{\partial x_1} + \frac{\partial N_{23}}{\partial x_2} + f_3 - \rho h \ddot{u}_3 = 0$$

- Equilibrium of moments:

$$\frac{\partial M_{11}}{\partial x_1} + \frac{\partial M_{12}}{\partial x_2} - N_{13} = 0$$

$$\frac{\partial M_{22}}{\partial x_2} + \frac{\partial M_{12}}{\partial x_1} - N_{23} = 0$$

Combining the three equations above, we obtain the **dynamic equilibrium equation for Kirchhoff-Love plate bending**:

$$\nabla_k^T \mathbf{M} + f_3 = \rho h \ddot{u}_3$$

Strong form for Kirchhoff-Love plate bending

Let $\Omega = [-a, a] \times [-b, b]$. Find the transverse displacement $u_3 \in C^4(\Omega \times [0, T])$ such that

$$\frac{h^3}{12} \nabla_k^T \mathbf{C} \nabla_k u_3 + \rho h \ddot{u}_3 = f_3 \quad \text{on } \Omega \times]0, T[$$

Boundary conditions (*simply supported*):

$$u_3 = 0 \quad \text{in } \partial\Omega \times]0, T[$$

$$\mathbf{M}_n = 0 \quad \text{in } \partial\Omega \times]0, T[$$

Initial conditions:

$$u_3(\cdot, 0) = u_0 \quad \text{in } \Omega$$

$$\dot{u}_3(\cdot, 0) = v_0 \quad \text{in } \Omega$$

In case of isotropic material, the strong form equation reduces to

$$D \left(\frac{\partial^4 u_3}{\partial x_1^4} + 2 \frac{\partial^4 u_3}{\partial x_1^2 \partial x_2^2} + \frac{\partial^4 u_3}{\partial x_2^4} \right) + \rho h \ddot{u}_3 = f_3$$

where $D = Eh^3/(12(1 - \nu^2))$.

Approximated boundary conditions

Simply supported on all 4 edges:

- No vertical displacement:

$$u_3 = 0 \quad \text{in } \partial\Omega \times]0, T[$$

- No moment resistance (free to rotate):

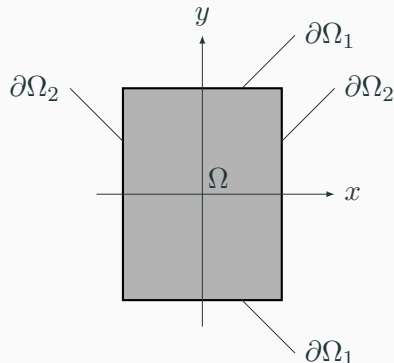
$$M_{11} = -D(\partial_x \varphi_2 + \nu \partial_y \varphi_1) = 0 \quad \text{in } \partial\Omega_1 \times]0, T[$$

$$M_{22} = -D(\nu \partial_x \varphi_2 + \partial_y \varphi_1) = 0 \quad \text{in } \partial\Omega_2 \times]0, T[$$

These conditions are replaced by the approximated conditions:

$$\varphi_2 = -\partial_x u_3 = 0 \quad \text{in } \partial\Omega_1 \times]0, T[$$

$$\varphi_1 = \partial_y u_3 = 0 \quad \text{in } \partial\Omega_2 \times]0, T[$$



Weak form for Kirchhoff-Love simply supported plate

The weak form consists of finding the transverse displacement $u_3 \in \mathcal{U}$ such that the following equation is satisfied for every virtual transversal displacement $\delta u_3 \in \mathcal{V}$:

$$\frac{h^3}{12} \int_{\Omega} (\nabla_k \delta u_3)^T \mathbf{C} \nabla_k u_3 d\Omega + \int_{\Omega} \rho h \ddot{u}_3 \delta u_3 d\Omega = \int_{\Omega} f_3 \delta u_3 d\Omega$$

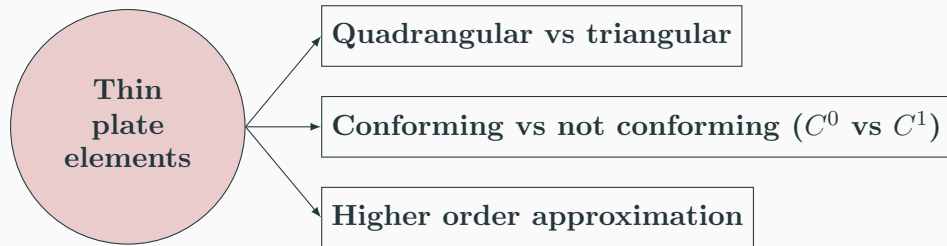
The functional spaces are defined as

$$\begin{aligned}\mathcal{U} &= \{u_3(\cdot, t) \in H^2(\Omega) \mid u_3 = 0 \text{ in } \partial\Omega \times]0, T[\} \\ \mathcal{V} &= \{\delta u_3 \in H^2(\Omega) \mid \delta u_3 = 0 \text{ in } \partial\Omega\}\end{aligned}$$

Thin plate bending elements

Overview of thin plate bending elements

Numerous finite elements for plate bending have been developed: **more than 88 distinct types** can be identified.



- **Adini-Melosh-Clough element (AMC):**
12 DOFs quadrangular, not conforming, thin plate bending element.
- **Crouzeix-Raviart (CR):**
16 DOFs quadrangular, conforming, thin plate bending element.

Finite element approximation of Kirchhoff-Love plate

- Weak form equation

$$\frac{h^3}{12} \int_{\Omega} (\nabla_k \delta u_3)^T \mathbf{C} \nabla_k u_3 d\Omega + \int_{\Omega} \rho h \ddot{u}_3 \delta u_3 d\Omega = \int_{\Omega} f_3 \delta u_3 d\Omega$$

- Semi-discrete weak form

$$\mathbf{M}\ddot{\mathbf{q}}(t) + \mathbf{K}\mathbf{q}(t) = \mathbf{r}(t)$$

Finite element approximation using polynomial shape functions:

$${}^e u_3^h(x_1, x_2, t) = \sum_{i=1}^n {}^e \mathbf{h}_i(x_1, x_2) {}^e \mathbf{q}^i(t) = {}^e \mathbf{H}(\mathbf{x}) {}^e \mathbf{q}(t)$$

Selection of the displacement approximation function

To ensure convergence it is necessary to consider:

Completeness criterion:

highest order of derivatives in the weak form is 2.



Complete polynomials of at least degree 2:

$$\begin{aligned} {}^e u_3^h = & a_1 + a_2 x + a_3 y + \\ & + a_4 x^2 + a_5 xy + a_6 y^2 + \\ & + \textit{higher order terms} \end{aligned}$$

Continuity criterion:

${}^e u_3^h$, $\partial_{x_1} {}^e u_3^h$, $\partial_{x_2} {}^e u_3^h$ are continuous between elements.



Each node i has 3 DOFs:

- ${}^e d^i$ vertical displacement,
- ${}^e \theta_1^i, {}^e \theta_2^i$ rotations.

AMC plate bending elements

Selection of the displacement approximation function

			1				Constant
			x_1		x_2		Linear
	x_1^2		x_1x_2		x_2^2		Quadratic
x_1^3		$x_1^2x_2$		$x_1x_2^2$		x_2^3	Cubic
x_1^4	$x_1^3x_2$		$x_1^2x_2^2$		$x_1x_2^3$	x_2^4	Quartic

Displacement approximation for rectangular elements with four nodes and thus 12 DOFs: complete cubic polynomial, augmented with two (*geometrically invariant*) quartic terms

$${}^e u_3^h(x_1, x_2, t) = a_1 + a_2x_1 + a_3x_2 + a_4x_1^2 + a_5x_1x_2 + a_6x_2^2 + \\ + a_7x_1^3 + a_8x_1^2x_2 + a_9x_1x_2^2 + a_{10}x_2^3 + a_{11}x_1^3x_2 + a_{12}x_1x_2^3$$

- ✓ Displacement approximation solves the unloaded strong form equation.
- ✓ Continuity in displacement along the interfaces of the elements.
- ✗ Slopes continuity along the interfaces are not ensured (*not conforming*).

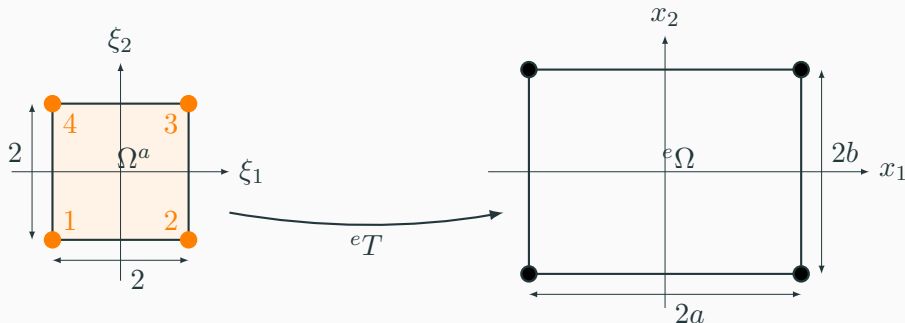
Coordinate transform

The coordinate transformation:

$${}^eT : \Omega^a \rightarrow {}^e\Omega$$

$$\boldsymbol{\xi} = \{\xi_1, \xi_2\}^T \mapsto \mathbf{x}(\boldsymbol{\xi}) = \{x_1(\boldsymbol{\xi}), x_2(\boldsymbol{\xi})\}^T = \{a\xi_1, b\xi_2\}^T$$

maps any point $\boldsymbol{\xi}$ in $\Omega^a = [-1, 1] \times [-1, 1]$ to its corresponding point of coordinate $\mathbf{x}(\boldsymbol{\xi})$ in ${}^e\Omega = [-a, a] \times [-b, b]$:



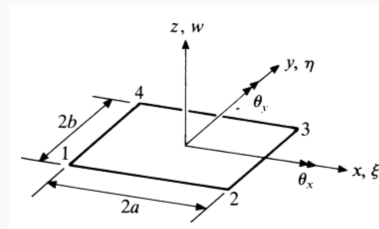
Approximate displacement in local coordinates

$$\begin{aligned}
 {}^e u_3^h(\boldsymbol{\xi}, t) &= \sum_{i=1}^4 {}^a \mathbf{h}_i(\boldsymbol{\xi}) {}^e \mathbf{q}^i(t) = {}^a \mathbf{H}(\boldsymbol{\xi}) {}^e \mathbf{q}(t) \\
 &= \begin{bmatrix} {}^a \mathbf{h}_1(\boldsymbol{\xi}) & {}^a \mathbf{h}_2(\boldsymbol{\xi}) & {}^a \mathbf{h}_3(\boldsymbol{\xi}) & {}^a \mathbf{h}_4(\boldsymbol{\xi}) \end{bmatrix} \begin{bmatrix} {}^e \mathbf{q}^1(t) \\ {}^e \mathbf{q}^2(t) \\ {}^e \mathbf{q}^3(t) \\ {}^e \mathbf{q}^4(t) \end{bmatrix}
 \end{aligned}$$

where

$$\blacksquare \quad {}^e \mathbf{q}^i(t) = \begin{bmatrix} {}^e d^i(t) \\ {}^e \theta_1^i(t) \\ {}^e \theta_2^i(t) \end{bmatrix} = \begin{bmatrix} {}^e u_3^h(\boldsymbol{\xi}^i, t) \\ \partial_{\xi_2} {}^e u_3^h(\boldsymbol{\xi}^i, t)/b \\ -\partial_{\xi_1} {}^e u_3^h(\boldsymbol{\xi}^i, t)/a \end{bmatrix}$$

■ $\boldsymbol{\xi}^i$ are the local coordinates of node i .



DOFs in local coordinates

- Assumed form of the displacement function:

$$\begin{bmatrix} {}^e u_3^h \\ \partial_{\xi_2} {}^e u_3^h \\ \partial_{\xi_1} {}^e u_3^h \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & \xi_1 & \xi_2 & \xi_1^2 & \xi_1 \xi_2 & \xi_2^2 & \xi_1^3 & \xi_1^2 \xi_2 & \xi_1 \xi_2^2 & \xi_2^3 & \xi_1^3 \xi_2 & \xi_1 \xi_2^3 \\ 0 & 0 & 1 & 0 & \xi_1 & 2\xi_2 & 0 & \xi_1^2 & 2\xi_1 \xi_2 & 3\xi_2^2 & \xi_1^3 & 3\xi_1 \xi_2^2 \\ 0 & 1 & 0 & 2\xi_1 & \xi_2 & 0 & 3\xi_1^2 & 2\xi_1 \xi_2 & \xi_2^2 & 0 & 3\xi_1^2 \xi_2 & \xi_2^3 \end{bmatrix}}_{\mathbf{P}(\xi)} \underbrace{\begin{bmatrix} \alpha_1(t) \\ \vdots \\ \alpha_{12}(t) \end{bmatrix}}_{\boldsymbol{\alpha}(t)}$$

- Then the DOFs in local coordinates are:

$${}^e \mathbf{q}^i(t) = \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1/b & 0 \\ 0 & 0 & -1/a \end{bmatrix} \mathbf{P}(\xi^i)}_{\bar{\mathbf{P}}(\xi^i)} \boldsymbol{\alpha}(t)$$

Shape functions matrix for the AMC element

- Since

$${}^e\mathbf{q}(t) = \begin{bmatrix} {}^e\mathbf{q}^1(t) \\ {}^e\mathbf{q}^2(t) \\ {}^e\mathbf{q}^3(t) \\ {}^e\mathbf{q}^4(t) \end{bmatrix} = \underbrace{\begin{bmatrix} \bar{\mathbf{P}}(\xi^1) \\ \bar{\mathbf{P}}(\xi^2) \\ \bar{\mathbf{P}}(\xi^3) \\ \bar{\mathbf{P}}(\xi^4) \end{bmatrix}}_{\mathbf{A}} \boldsymbol{\alpha}(t) \quad \Rightarrow \quad \boldsymbol{\alpha}(t) = \mathbf{A}^{-1} {}^e\mathbf{q}(t)$$

- Then the shape functions matrix is ${}^a\mathbf{H} = [{}^a\mathbf{h}_1 \quad {}^a\mathbf{h}_2 \quad {}^a\mathbf{h}_3 \quad {}^a\mathbf{h}_4] = \mathbf{P}_1 \mathbf{A}^{-1}$ where

$${}^a\mathbf{h}_i(\boldsymbol{\xi}) = \begin{bmatrix} (1 + \xi_1^i \xi_1)(1 + \xi_2^i \xi_2)(2 + \xi_1^i \xi_1 + \xi_2^i \xi_2 - \xi_1^2 - \xi_2^2)/8 \\ b(1 + \xi_1^i \xi_1)(\xi_2^i + \xi_2)(\xi_2^2 - 1)/8 \\ -a(\xi_1^i + \xi_1)(\xi_1^2 - 1)(1 + \xi_2^i \xi_2)/8 \end{bmatrix}^T$$

and $\boldsymbol{\xi}^i = \{\xi_1^i, \xi_2^i\}^T$ are the local coordinates of node i .

Deformation, stiffness, mass matrices and loads vector

- ${}^a\mathbf{B} = \nabla_k {}^a\mathbf{H}$ is a (3×12) matrix:

$${}^a\mathbf{B} = \begin{bmatrix} \frac{1}{a^2} \partial_{\xi_1}^2 \\ \frac{1}{b^2} \partial_{\xi_2}^2 \\ \frac{2}{ab} \partial_{\xi_1 \xi_2}^2 \end{bmatrix} \begin{bmatrix} {}^a\mathbf{h}_1 & {}^a\mathbf{h}_2 & {}^a\mathbf{h}_3 & {}^a\mathbf{h}_4 \end{bmatrix} = \begin{bmatrix} \begin{bmatrix} \frac{1}{a^2} \partial_{\xi_1}^2 \\ \frac{1}{b^2} \partial_{\xi_2}^2 \\ \frac{2}{ab} \partial_{\xi_1 \xi_2}^2 \end{bmatrix} {}^a\mathbf{h}_1 & \dots & \begin{bmatrix} \frac{1}{a^2} \partial_{\xi_1}^2 \\ \frac{1}{b^2} \partial_{\xi_2}^2 \\ \frac{2}{ab} \partial_{\xi_1 \xi_2}^2 \end{bmatrix} {}^a\mathbf{h}_4 \end{bmatrix}$$

- ${}^e\mathbf{K}$ and ${}^e\mathbf{M}$ are (12×12) matrices and ${}^e\mathbf{r}$ is a (12×1) vector:

$${}^e\mathbf{K} = \frac{h^3}{12} \int_{\Omega} {}^a\mathbf{B}^T \mathbf{C}^a \mathbf{B} d\Omega = \frac{h^3 ab}{12} \int_{-1}^1 \int_{-1}^1 {}^a\mathbf{B}^T \mathbf{C}^a \mathbf{B} d\xi_1 d\xi_2,$$

$${}^e\mathbf{M} = \int_{\Omega} \rho h {}^a\mathbf{H}^T {}^a\mathbf{H} d\Omega = \rho h ab \int_{-1}^1 \int_{-1}^1 {}^a\mathbf{H}^T {}^a\mathbf{H} d\xi_1 d\xi_2,$$

$$\mathbf{r}(t) = \int_{\Omega} {}^a\mathbf{H}^T f_3(t) d\Omega = ab f_3(t) \int_{-1}^1 \int_{-1}^1 {}^a\mathbf{H}^T d\xi_1 d\xi_2.$$

- The **approximated stresses** ${}^e\sigma_{11}^h$ and ${}^e\sigma_{22}^h$ and ${}^e\sigma_{12}^h$ at any point (x_1, x_2, x_3) of the element e are given in term of the nodal displacements:

$${}^e\boldsymbol{\sigma}^h = \mathbf{C}^e \boldsymbol{\varepsilon}^h = -x_3 \mathbf{C}^e \nabla_k {}^e u_3^h = -x_3 \mathbf{C}^e \mathbf{B}^e \mathbf{q}(t)$$

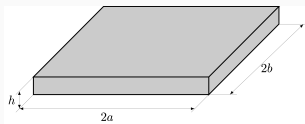
- The **approximated bending moments** ${}^e M_{11}^h$ and ${}^e M_{22}^h$ and twisting moment ${}^e M_{12}^h$ per unit length are given by:

$${}^e \mathbf{M} = \begin{bmatrix} {}^e M_{11}^h \\ {}^e M_{22}^h \\ {}^e M_{12}^h \end{bmatrix} = \int_{-\frac{h}{2}}^{\frac{h}{2}} x_3 \begin{bmatrix} {}^e \sigma_{11}^h \\ {}^e \sigma_{22}^h \\ {}^e \sigma_{12}^h \end{bmatrix} dx_3 = -\frac{h^3}{12} \mathbf{C}^e \mathbf{B}^e \mathbf{q}(t)$$

Example: isotropic square plate in free vibrations

Example: isotropic square plate in free vibrations

Use AMC elements to estimate the five lowest frequencies of an isotropic square plate ($\ell \times \ell$) which is simply supported on all four edges (SSSS).



Compare the results with the analytical solution

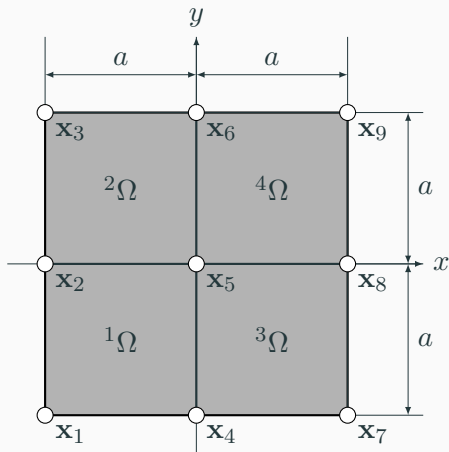
$$\omega_{m,n} = \pi^2 \frac{m^2 + n^2}{\ell^2} \sqrt{\frac{D}{\rho h}} \text{ rad/s},$$

where ℓ is the length of each side and (m, n) are the number of half-waves in the x - and y -directions and D is flexural rigidity of the plate:

$$D = \frac{Eh^3}{12(1 - \nu^2)}.$$

Simply supported isotropic square plate

Discretization with a rectangular mesh of 2×2 AMC elements.



- $2a = 1$ length [m]
- $2a = 1$ height [m]
- $h = 0.01$ thickness [m]
- $E = 210$ Young's modulus [GPa]
- $\nu = 0.3$ Poisson's ratio
- $\rho = 7800$ material density [kg/m^3]

Symmetry and boundary conditions

We exploit the symmetry of the problem by modeling only one-quarter of the plate with 4 AMC elements, applying appropriate boundary conditions to account for the symmetric and antisymmetric modes along the edges.

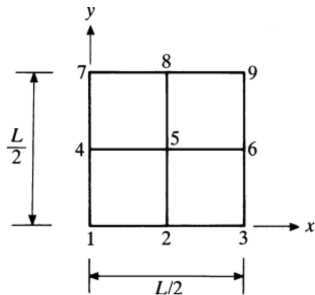


Figure 1: One-quarter of the plate represented by four rectangular elements.

■ Simple support:

- sides 1-3: $u_3 = \theta_2 = 0$ at nodes 1, 2, and 3,
- sides 1-7: $u_3 = \theta_1 = 0$ at nodes 1, 4, and 7.

■ Symmetric modes:

- with respect to side 3-9: $\theta_2 = 0$ at nodes 3, 6, and 9,
- with respect to side 7-9: $\theta_1 = 0$ at nodes 7, 8, and 9.

■ Antisymmetric modes:

- with respect to side 3-9: $u_3 = \theta_1 = 0$ at nodes 3, 6, and 9.
- with respect to side 7-9: $u_3 = \theta_2 = 0$ at nodes 7, 8, and 9.

Frequencies approximation for square plate with AMC elements

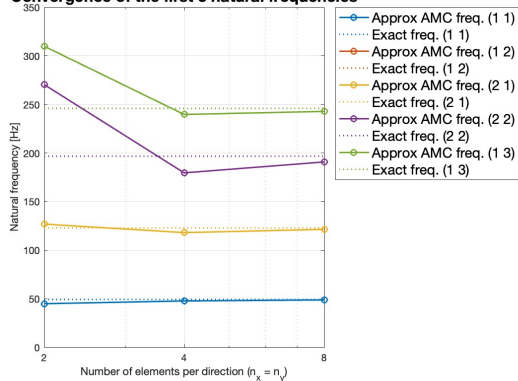
Comparison of exact and approximate natural frequencies with a mesh of 2×2 AMC elements for a thin SSSS square plate.

m	n	Exact Freq. [Hz]	Approx. Freq. [Hz]	Rel. Error (%)
1	1	49.171	44.892	-8.7032
1	2	122.93	126.82	3.1687
2	1	122.93	126.82	3.1687
2	2	196.69	270.44	37.496
1	3	245.86	309.92	26.059
3	1	245.86	517.41	110.45
2	3	319.61	517.41	61.886

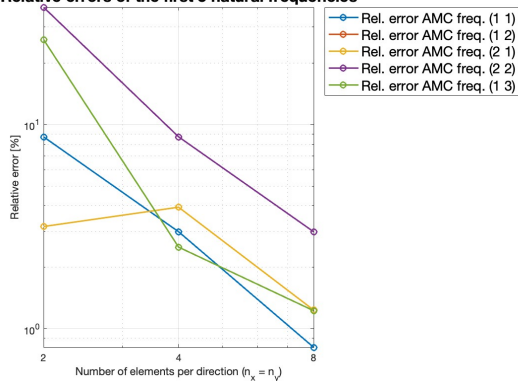
Frequencies approximation for square plate with AMC elements

In these figures, we present the convergence and relative errors of the first five natural frequencies of a square plate, obtained using AMC elements for different mesh refinements.

Convergence of the first 5 natural frequencies



Relative errors of the first 5 natural frequencies



Modal shapes

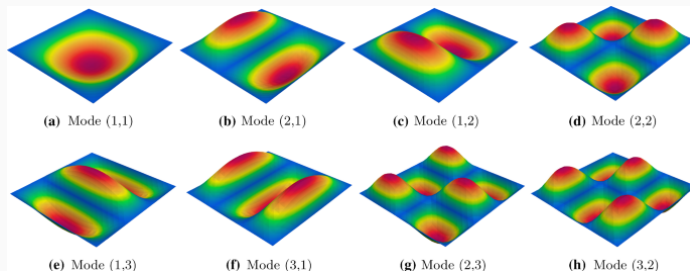
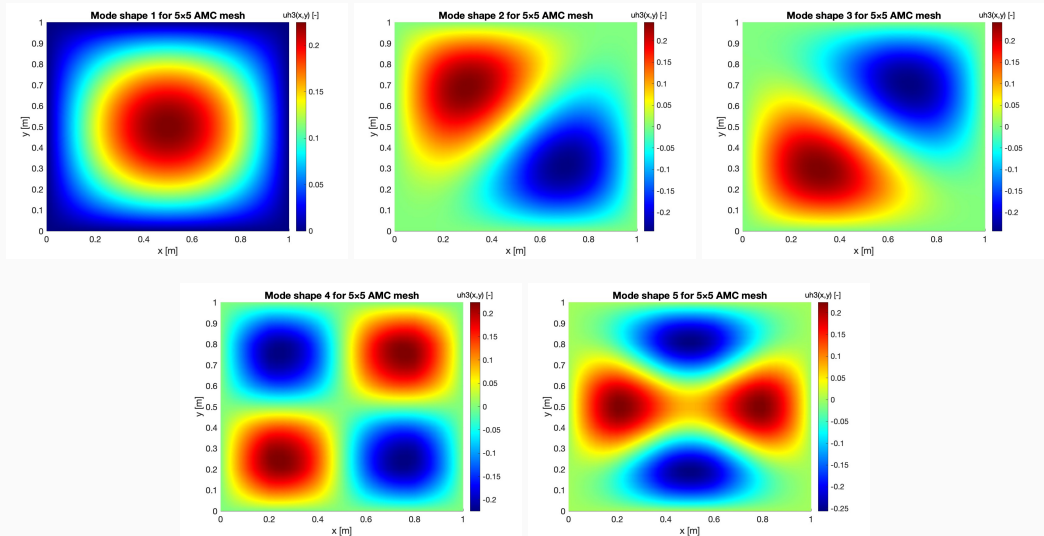


Figure 2: Mode shapes for a simply supported (SSSS) square plate.

(Credit: Pagani, Azzara, Carrera - Geometrically nonlinear analysis and vibration of in-plane-loaded variable angle tow composite plates and shells)

Modal patterns



CR thin plate bending elements

Selection of the displacement approximation function

1	Constant
x_1	Linear
x_2	Linear
x_1^2	Quadratic
x_1x_2	Quadratic
x_2^2	Quadratic
x_1^3	Cubic
$x_1^2x_2$	Cubic
$x_1x_2^2$	Cubic
x_2^3	Cubic
$x_1^3x_2$	Quartic
$x_1^2x_2^2$	Quartic
$x_1x_2^3$	Quartic
$x_1^3x_2^2$	Quintic
$x_1^2x_2^3$	Quintic
$x_1^3x_2^3$	Sextic

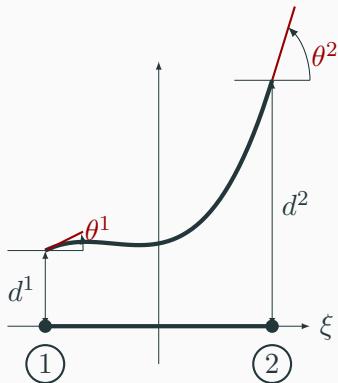
Displacement approximation for rectangular elements with four nodes and 16 DOFs (4 DOFs per node): complete cubic polynomial, augmented with three quartic, two quintic and one sextic terms:

$$\begin{aligned} {}^e u_3^h(x_1, x_2, t) = & a_1 + a_2x_1 + a_3x_2 + a_4x_1^2 + a_5x_1x_2 + a_6x_2^2 + \\ & + a_7x_1^3 + a_8x_1^2x_2 + a_9x_1x_2^2 + a_{10}x_2^3 + \\ & + a_{11}x_1^3x_2 + a_{12}x_1x_2^3 + a_{13}x_1^2x_2^2 + a_{14}x_1x_2^3 + a_{15}x_1^3x_2^2 + a_{16}x_1^3x_2^3 \end{aligned}$$

Recall: transversal bending for Euler-Bernoulli beam

Approximated transversal displacement u_2^h as a function of the nodal DOFs:

$${}^e u_2^h(\xi, t) = \mathbf{H}(\xi) \mathbf{q}(t) = [h_1(\xi), h_2(\xi), h_3(\xi), h_4(\xi)] \begin{bmatrix} d^1(t) \\ \theta^1(t) \\ d^2(t) \\ \theta^2(t) \end{bmatrix}$$



C^1 Hermite shape functions on $[-1, 1]$:

$$h_1(\xi) = (\xi^3 - 3\xi + 2)/4$$

$$h_3(\xi) = (-\xi^3 + 3\xi + 2)/4$$

$$h_2(\xi) = (\xi^3 - \xi^2 - \xi + 1)/4$$

$$h_4(\xi) = (\xi^3 + \xi^2 - \xi - 1)/4$$

Shape functions matrix: a first try

- Let ξ^i be the coordinate of the node i . Then the Hermite shape functions matrix is $\mathbf{H}(\xi) = [f_1 \ g_1 \ f_2 \ g_2]$ where

$$f_i(\xi) = (-\xi^i \xi^3 + 3\xi^i \xi + 2)/4 \qquad g_i(\xi) = (\xi^3 + \xi^i \xi^2 - \xi - \xi^i)/4.$$

- The shape functions matrix for the plate bending element is a product of Hermite functions:

$${}^a\mathbf{H} = [{}^a\mathbf{h}_1 \quad {}^a\mathbf{h}_2 \quad {}^a\mathbf{h}_3 \quad {}^a\mathbf{h}_4]$$

where

$${}^a\mathbf{h}_i(\boldsymbol{\xi}) = \begin{bmatrix} f_i(\xi_1)f_i(\xi_2) \\ b f_i(\xi_1)g_i(\xi_2) \\ -a g_i(\xi_1)f_i(\xi_2) \end{bmatrix}^T.$$

Zero-twist constraint

- The approximate displacement in local coordinate is defined as:

$${}^e u_3^h(\boldsymbol{\xi}, t) = \sum_{i=1}^4 {}^a \mathbf{h}_i(\xi_1, \xi_2) {}^e \mathbf{q}^i(t) = {}^a \mathbf{H}(\boldsymbol{\xi}) {}^e \mathbf{q}(t)$$

- ✗ The twist

$$\frac{\partial^2}{\partial \xi_1 \partial \xi_2} {}^e u_3^h$$

is zero at the four nodal points. Thus the plate will tend to a zero twist condition as an increasing number of elements are used.

Solution: introduce twist

$$\theta_{12}^i = \frac{\partial^2}{\partial \xi_1 \partial \xi_2} {}^e u_3^h(\boldsymbol{\xi}^i, t)$$

as an extra degree of freedom.

Shape functions matrix for the CR element

The approximate displacement in local coordinate is defined as:

$${}^e u_3^h(\boldsymbol{\xi}, t) = \sum_{i=1}^4 {}^a \mathbf{h}_i(\boldsymbol{\xi}) {}^e \mathbf{q}^i(t) = {}^a \mathbf{H}(\boldsymbol{\xi}) {}^e \mathbf{q}(t)$$

■ 4 DOFs per node: ${}^e \mathbf{q}^i(t) = \begin{bmatrix} {}^e d^i(t) \\ {}^e \theta_1^i(t) \\ {}^e \theta_2^i(t) \\ {}^e \theta_{12}^i(t) \end{bmatrix} = \begin{bmatrix} {}^e u_3^h(\boldsymbol{\xi}^i, t) \\ \partial_{\xi_2} {}^e u_3^h(\boldsymbol{\xi}^i, t)/b \\ -\partial_{\xi_1} {}^e u_3^h(\boldsymbol{\xi}^i, t)/a \\ \partial_{\xi_1 \xi_2}^2 {}^e u_3^h(\boldsymbol{\xi}^i, t)/(ab) \end{bmatrix}$

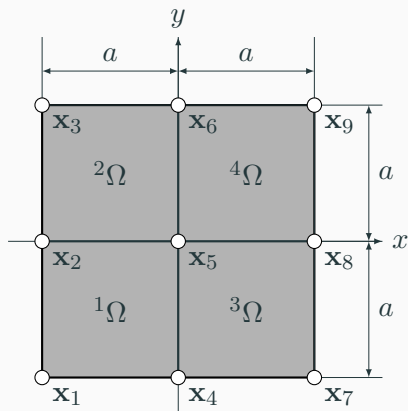
■ The shape function matrix is ${}^a \mathbf{H} = [{}^a \mathbf{h}_1 \quad {}^a \mathbf{h}_2 \quad {}^a \mathbf{h}_3 \quad {}^a \mathbf{h}_4]$ where

$${}^a \mathbf{h}_i(\boldsymbol{\xi}) = \begin{bmatrix} f_i(\xi_1) f_i(\xi_2) \\ b f_i(\xi_1) g_i(\xi_2) \\ -a g_i(\xi_1) f_i(\xi_2) \\ a b g_i(\xi_1) g_i(\xi_2) \end{bmatrix}^T$$

Example: isotropic square plate in free vibrations

Frequencies estimate for square plate with CR element

A mesh of 2×2 CR elements is used to estimate the fundamental frequencies of a square plate ($\ell \times \ell$) which is simply supported on all four edges (SSSS).



- $2a = 1$ length $[m]$
- $2a = 1$ height $[m]$
- $h = 0.01$ thickness $[m]$
- $E = 210e9$ Young's modulus $[Pa]$
- $\nu = 0.3$ Poisson's ratio
- $\rho = 7800$ material density $[kg/m^3]$

Frequencies approximation for square plate with CR elements

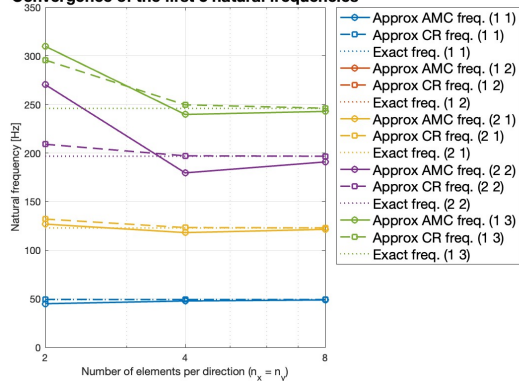
Comparison of exact and approximate natural frequencies with a mesh of 2×2 CR elements for a thin SSSS square plate.

m	n	Exact Freq. [Hz]	Approx. Freq. [Hz]	Rel. Error (%)
1	1	49.171	49.277	0.21558
1	2	122.93	132	7.3816
2	1	122.93	132	7.3816
2	2	196.69	209.01	6.2666
1	3	245.86	295.69	20.269
3	1	245.86	295.69	20.27
2	3	319.61	366.03	14.523

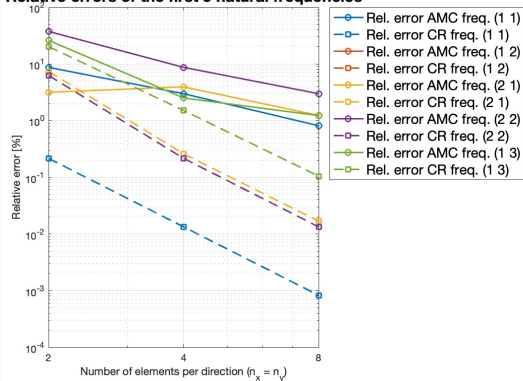
Comparison: AMC vs CR plate bending element

In these figures, we present the convergence and relative errors of the first five natural frequencies of a square plate, obtained using AMC and CR elements for different mesh refinements.

Convergence of the first 5 natural frequencies



Relative errors of the first 5 natural frequencies



Comparison: AMC vs CR plate bending element

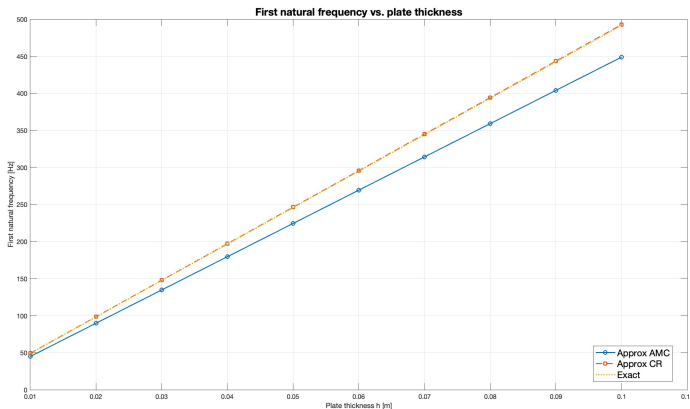


Figure 3: Evolution of the approximated and exact fundamental frequencies with thickness for a SSSS square plate.

Comparison: AMC vs CR plate bending element

AMC element (*Non conforming*)

- 3 degrees of freedom per node: u_3 , θ_1 , θ_2 .
- Only displacement u_3 is continuous across elements; rotations may have jumps.
- Nonconforming element: does not fully satisfy C^1 continuity.
- Simpler and computationally cheaper; suitable for practical applications.

CR element (*Conforming*)

- 4 degrees of freedom per node: u_3 , θ_1 , θ_2 , and θ_{12} .
- Displacement u_3 and rotations θ_1, θ_2 are continuous across element boundaries.
- Fully conforming to the C^1 continuity required by Kirchhoff plate theory.
- Higher computational cost and complexity.

Why would you use AMC element if it's nonconforming?

- Computationally cheaper than full C^1 elements (fewer DOFs).
- It still converges (theoretical results for nonconforming FEMs show convergence *under certain conditions*).
- Suitable when small slope discontinuities are acceptable (e.g., dynamic problems, large meshes).
- Tends to underestimate the natural frequencies, making it a useful benchmark for detecting overstiffness in numerical model.